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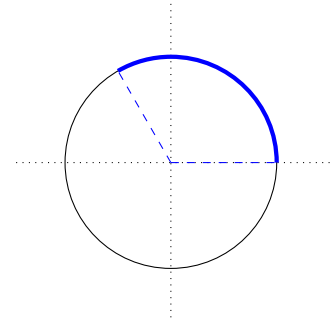
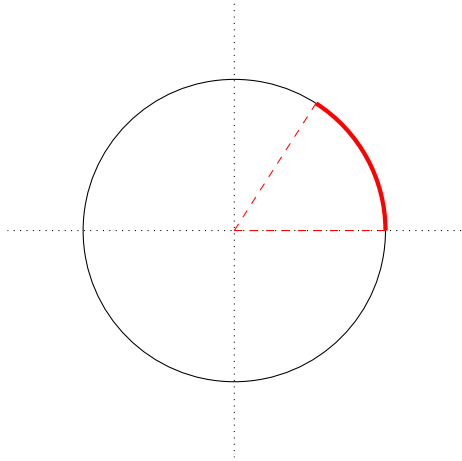
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Rotational Motion Notes

Angular Quantities

Measuring Angles



Conversions

Angular Kinematic Quantities

Concept	Linear/Translational Quantity	Angular/Rotational Quantity	“Bridge”
position			
displacement			
	units:	units:	
velocity			
	units:	units:	
acceleration			
	units:	units:	

1. A bicycle has wheels of diameter 68 cm. If the bicycle travels 92 km, how many rotations do the wheels make?
2. A wheel of radius 1.3 meters is accelerating at a rate of 12 rad/s^2 . At the moment that the wheel is rotating at 3.2 rad/s ,
 - (a) what is the magnitude of *tangential* acceleration at a point on the outside of the wheel?
 - (b) what is the magnitude of *radial* (centripetal) acceleration at a point on the outside of the wheel?
 - (c) what is the magnitude of the *total* acceleration at a point on the outside of the wheel?

Rotational Kinematic Equations

“Old Faithful”

$$v = v_0 + at$$

“The Big Chalupa”

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

“Ain’t Got No Time”

$$v^2 = v_0^2 + 2a(x - x_0)$$

Practice

Pilots can be tested for the stresses of flying high-speed jets in a whirling “human centrifuge,” which starts at rest and takes 1.0 min to turn through 20 complete revolutions before reaching its final speed.

- (a) What was its angular acceleration (assumed constant)?
- (b) What was its final angular speed **in rpm**?

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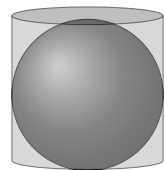
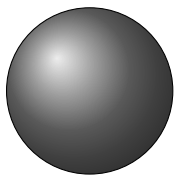
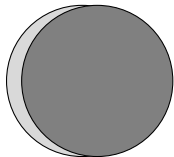
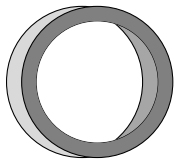
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Rotational Dynamics

Quantities

Concept	Linear/Translational Quantity	Angular/Rotational Quantity	“Bridge”
cause of acceleration			
	units:	units:	
inertia			
	units:	units:	

Moment of Inertia for Extended Objects



Practice

1. Consider the following situation:

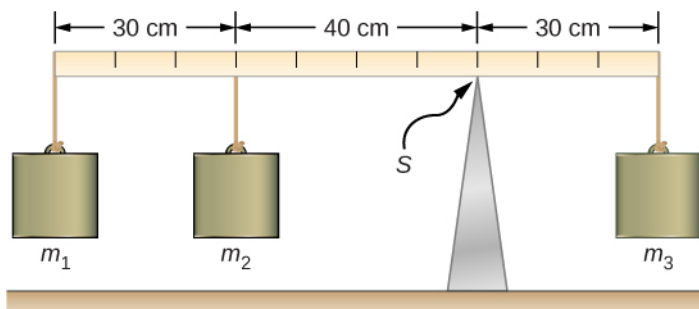


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- (a) Calculate the net torque about point S if $m_1 = 10$ kg, $m_2 = 20$ kg, and $m_3 = 30$ kg.
- (b) If you took m_3 off, what mass could you replace it with such that the system would balance?

2. A torque of 1.20 m N is applied to a disk of mass 4.80 kg and radius 30 cm until it reaches a rotational speed of 10,300 rpm. Through how many revolutions does the disk rotate during this process?

Rotational Energy

Quantities

Concept	Linear/Translational Quantity	Angular/Rotational Quantity	“Bridge”
kinetic energy			
	units:	units:	
work			
	units:	units:	

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Practice

1. A 7-kg solid ball of diameter 30 cm rolls without slipping at a velocity of 3.5 m/s. Calculate the total kinetic energy of the ball.
2. Consider a solid disc that is released from rest at the top of a ramp that is 5 meters tall. Assume that the disc does not slip and is perfectly round. How fast will the disc be going when it gets to the bottom of the ramp?

Angular Momentum

Quantities

Concept	Linear/Translational Quantity	Angular/Rotational Quantity	“Bridge”
momentum			
	units:	units:	



1. An ice skater is spinning with her arms extended. She has a moment of inertia of 4.1 kg m^2 and an initial angular velocity of 2.0 rad/s . She then pulls her arms in, reducing her moment of inertia to 1.5 kg m^2 .
 - (a) What is her new angular velocity after pulling her arms in?
 - (b) Calculate her initial and final rotational kinetic energy. Has her rotational kinetic energy increased or decreased? If so, explain why.
2. The bicycle wheel is essentially a hoop of mass 8 kg and radius 0.4 m . The turntable is essentially a disc of mass 5 kg and radius 0.5 m . The instructor is essentially a particle located at the center of the axis. If the bicycle wheel has an initial angular speed of 4.3 rad/s , what is the rate of rotation of the instructor and turntable after the wheel is flipped?